

Manoj Chapagain

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SUMMARY STATEMENT

Experienced software engineer with 8 years in web development, distributed systems, and cybersecurity. Proven track record building and leading teams to deliver high-quality products on tight deadlines across fintech, cybersecurity, IoT, and SaaS sectors.

EXPERIENCE

Full-Stack Developer

Feb. 2024 – Present

Mechatronic

Canada

- Led end-to-end development across frontend, backend, CI/CD pipelines, and production infrastructure using DevOps practices
- Developed a firmware module for IoT devices and built the user-facing web interface
- Developed AI agents, tools, and automations for efficient and scalable workflows

Backend Developer

May. 2022 – Apr. 2023

Microsec

Singapore

- Developed lightweight PKI-based security protocols in Python for IoT/OT devices using the Cryptography library; enabled certificate provisioning, encrypted messaging, and integrity checks for resource-constrained environments
- Architected and deployed microservices with Docker and Kubernetes, orchestrating services across a multi-node cluster to ensure high availability and scalability

Sr. Software Engineer

Jan. 2021 – Mar. 2022

Codavatar

Kathmandu, Nepal

- Led the backend team in delivering scalable, production-ready services under tight deadlines, ensuring on-time delivery of key milestones
- Developed and optimized core features for Krispcall, a cloud-based phone system leveraging WebRTC and Twilio
- Adopted domain-driven and clean architecture principles, applying design patterns to enhance code quality and maintainability
- Performed performance tuning and refactoring to improve system scalability, reliability, and response times

Full Stack Developer

Nov. 2017 – Dec. 2020

Janaki Tech.

Lalitpur, Nepal

- Built the front-end from scratch using React, growing the user base from few thousand to 100K within one year for an early-stage fintech startup
- Engineered a payment gateway and digital wallet using Python, Django, React.js, and AWS; integrated with banks, telecoms, and e-commerce platforms via third-party APIs

Jr. Full-Stack Developer

Dec. 2016 – Nov. 2017

Grepsr

Kathmandu, Nepal

- Built and maintained web scraping and data-extraction pipelines using microservice architecture, gRPC, Docker, and Kubernetes

EDUCATION

Project Managemnt - IT

May. 2023 – Dec. 2023

Seneca College

Toronto, Canada

Bachelor's degree in Computer Engineering

Nov. 2012 – Sep. 2016

Tribhuvan University, IOE Pulchowk Campus

Lalitpur, Nepal

OPEN SOURCE PROJECTS

Employee Management System | *Python, Django, drf, React, PostgreSQL, Docker* | *Source*

June 2020

databases-extensions | *SqlAlchemy, Postgres, GraphQL, Cursor Pagination* | *Source*

May 2021 – Present

PUBLICATIONS

Effects of Auto-Tuning and Pitch Normalization on Query by Humming

June 2020

International Journal of Advanced engineering

Source

TECHNICAL SKILLS

Languages/Scripts: Python, JavaScript/TypeScript, SQL, Shell (sh, zsh, bash)

Frameworks/Libraries: FastAPI, Flask, Django, DRF, Starlette, gRPC, GraphQL (Ariadne, Graphene), SQLAlchemy, RabbitMQ, Redis, OpenGL, Pandas, Matplotlib, React, Node.js, Selenium, LangGraph, OpenAI API, Pinecone, scikit-learn

DevOps/Cloud: Docker, Kubernetes (K8s), Ansible, Nginx, CI/CD (GitHub Actions, GitLab CI), AWS (EC2, S3, RDS, Lambda), Terraform, Prometheus, Grafana

Testing/Docs: Pytest, Postman, Swagger/OpenAPI, MkDocs, Ruff, Flake8, Cypress

Others: Linux, Jira, Asana